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Project Title:	Acute and Long Term Outcomes of Simulated Deep Space Radiation Exposure on Latent Viral CNS Infection and CNS Pathology		
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Program/Discipline:	HUMAN RESEARCH		
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Human Research Program Elements:	(1) **SR**:Space Radiation		
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Space Biology Element:	None		
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Zip Code:	45229-3026	**Congressional District:**	1
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Thompson, Richard (University of Cincinnati) Williams, Michael (Children's Hospital Medical Center, Cincinnati)		
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Task Description:	The successful diaspora of humans into deep space requires an understanding of the risk of deep space radiation exposure on short and long term health outcomes, including risks to the central nervous system (CNS). Assessing CNS risk must consider the impact of deep space radiation on the behavior of latent neurotropic infectious agents that reside within the human CNS. Herpes simplex virus (HSV) is among the most widespread agent in this category, permanently infecting the nervous systems of more than 80% of the global population. Latent HSV cannot be eliminated from the nervous system and there is no vaccine to prevent infection. Thus for the foreseeable future, humans traveling into deep space will harbor this and other alpha-herpesviruses. While uncontrolled HSV replication in the CNS results in often fatal encephalitis, this virus also has the capacity to promote chronic inflammation and neurodegeneration. Indeed there is substantial evidence that in combination with genetic risk factors, HSV infection in the CNS further increases the probability of developing Alzheimer's disease. In this proposal, the effects of simulated deep space radiation on the development of CNS pathology (reactive lesions) and the impact of latent and induced in vivo reactivation of HSV infection on the development of CNS pathology will be determined together with the role that human ApoE alleles play in these processes. We will employ our well characterized mouse model of HSV latency and in vivo reactivation in combination with quantitative measures of reactive lesions in the CNS to determine the effect of simulated deep space radiation on HSV mediated CNS damage. The acute stage of infection will be monitored using our standard approaches including quantitative evaluation of infectious virus titers. Latent infection in the trigeminal ganglia and brain will be quantified at 30-40 days postinfection (pi) using our quantitative real time PCR assay for the viral genome. Spontaneous and induced exit from latency will be quantified at the single neuron level using our whole TG immunohistochemical assay. This approach permits us to identify even a single positive neuron in groups of ganglia. Viral reactivation (production of infectious virus) will be evaluated using our standard assay. Reactivation will be induced using our well-established hyperthermic stress procedure (temperature of the mice is raised to 42.5 C for 10 minutes). We have shown and published that this procedure reliably induces viral reactivation and can be repeatedly employed two times per week for periods of up to at least one year pi. We have determined the repeated reactivation stress of C57Bl/6 mice results in reactive astrocytic lesions in the CNS of HSV infected but not in uninfected mice. Interestingly this damage is much more extensive in mice that express the human ApoE4 allele or in IL10 knock out mice. This suggests important relevance to human biology since both IL10 polymorphisms and the ApoE4 allele have been shown to be associated with increased risk of acquiring Alzheimer's disease. We now propose to collaborate with radiation biologists at NASA to determine appropriate conditions to simulate deep space radiation and we will quantify the effects of such radiation on CNS damage induced by reactivation of HSV latent infections. Significance: The goal of NASA's space radiation research is to enable the human exploration of space within acceptable risks from space radiation. Importantly, the proposed studies will determine the integrated risk of the interaction of galactic cosmic radiation with uninfected and latently infected CNS. The prevalence of HSV infection among humans makes this a nearly unavoidable additional factor that could significantly heighten the risk of CNS damage upon GCR (galactic cosmic radiation) exposure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Importantly, beyond its significance to deep space exploration, to our knowledge, this project is the first to systematically examine the long term impact of HSV latent/reactivation cycles in the nervous system in an experimental model system. These studies will therefore pioneer a model that will be extremely useful in understanding the long term outcomes of latent HSV in the nervous system among humans on Earth. Understanding the consequences on long term latency/reactivation is the first step toward developing treatment and preventive strategies.
	Manned exploration into deep space will include exposure of astronauts to primary galactic cosmic radiation (GCR) and solar proton events (SPE). Possible acute and long term risks to the CNS from galactic cosmic rays (GCRs) and solar proton events (SPEs) are a documented concern for human exploration of our solar system. The high energy and large mass of HZE nuclei (heavy nuclei >3) associated with GCR and SPE have the capacity to produce columns of damaged cells along their trajectory through the tissue (for review see (Cucinotta, Wang et al., 2008)). (Shukitt-Hale, Casadesus et al., 2000; Manda, Ueno et al., 2007; Manda, Ueno et al., 2008; Manda, Ueno et al., 2008; Villasana, Poage et al., 2008; Villasana, Pfankuch et al., 2010; Poulose, Bielinski et al., 2011; Britten, Davis et al., 2012)). One important yet understudied risk is the effect of GCR on latent infectious agents commonly present within the human CNS. Herpes simplex virus (HSV), a member of the human alpha herpesvirus family is among the most widespread agent in this category, permanently infecting the nervous systems of more than 80% of the global population (Whitley 2002). Latent HSV cannot be eliminated from the nervous system and there is no vaccine to prevent infection. Thus, for the foreseeable future, humans traveling into deep space will harbor this and potentially other alpha-herpesviruses in their nervous systems. The interacting pathways of stress and viral reactivation are well recognized. These combined with the potential for both immune suppression and heightened neuroinflammation following 56Fe exposure suggest that latent viral infections in the CNS are may pose an additional risk that must be evaluated. Data specific to the spaceflight environment must be compiled to quantify the magnitude of this risk. If this is identified as a risk of high enough magnitude then appropriate protection strategies should be employed. In this project, we are filling this gap using an advanced model of HSV latency and repetitive in vivo reactivation to assess the short and long term biological outcomes in the latently infected CNS following exposure to mission-relevant doses of HZE-particle radiation. Once feasibility and safety were demonstrated (year 1), tasks required for Experiment 2, the first of two long term experiments, were undertaken. Experiment 2 (12 month study) is coming to a close with respect to the animal studies which includes assessment of overall health, weight, an extensive battery of behavioral tests, and a pilot live animal MRI and spectroscopy. No differences (with the exception of eye disease) in the physical appearance, activity, or body weight of the animals among the groups-- latently infected, mock infected, irradiated or not irradiated-- were observed. The full analysis of the behavioral data is now completed revealing a correlation between long term HSV latent infection and behavioral deficit and reduced brain weight despite no loss in full body weight. This is the first study to look at the long term outcome of latent HSV in the CNS using behavioral tests. Radiation exposure did not significantly exacerbate the behavioral effects observed as a result of long term infection in these mice. The MR spectroscopy has also revealed differences in key metabolites pre and 10 months post radiation exposure, indicating that this will be a useful approach to monitor changes over time in the live animals. Blood, brain, and other tissues have been collected. The radiation exposed mice had spleen weights significantly reduced from unexposed mice at 12 months post irradiation. We asked whether this reduced spleen weight reflected a functional defect in the ability of the animals to (1) maintain effective neutralizing antibody titers to HSV, and (2) to prevent reinfection with HSV. Both of these aspects of immune function

Task Progress:	require the integrated networking of a competent immune system and would thus have the potential to reveal deficits due to radiation exposure. In both of these assays, the exposed mice were not different than unexposed controls, with full protection against reinfection and high neutralizing antibody titers. The biochemical and histologic analyses of these samples are in progress. An additional feature of the study was the development of HSV related eye disease. This aspect of the study revealed that repeated stress resulting in a significant increase in the number of animals developing herpetic stromal keratitis. This is a serious disease in humans which can lead to blindness and new insights into this disease process may result from these studies. Experiment 3 started in August, 2015. Animals for this experiment include mice carrying the human ApoE4 allele which is a potential risk factor for neurological disease development. These animals as well as control groups have been infected and progression to latency has occurred. Exposure to simulated GCR of Exp 3 took place during NASA Space Radiation Laboratory (NSRL)-15C. We followed up on an observation made during the initial pilot study and determined that proton-radiation exposure results in the HSV viral genome entry into the lytic cycle in the peripheral nervous system. This parallels reactivation induced by other known stimuli and in the immune suppressed context of spaceflight, may present a risk to crew health. We have now determined that latent infection of huApoE4 mice results in cognitive deficits, extending results from Exp 2. Our most recent findings indicate that host genetics could result in enhanced susceptibility to including HSV latent infection and radiation exposure. In summary, during this past 10 months, significant progress has been made toward completing the first long term experiment and progressing well into the second long term study. Importantly, these studies are revealing risks related to human health during space travel as well as to human health on Earth. References: Cucinotta, F. A., H. Wang, et al. (2008). "Risk of Acute or Late Central Nervous System Effects from Radiation Exposure." H NASA, HHC: 191-212. Shukitt-Hale, B., G. Casadesus, et al. (2000). "Spatial learning and memory deficits induced by ex-posure to iron -56-particle radiation." Radiat Res154(1): 28-33. Manda, K., M. Ueno, et al. (2008). "Memory impairment, oxidative damage and apoptosis induced by space radiation: ameliorative potential of alpha-lipoic acid." Behav Brain Res 187(2): 387-95. Manda, K., M. Ueno, et al. (2008). "Space radiation-induced inhibition of neurogenesis in the hippocampal dentate gyrus and memory impairment in mice: ameliorative potential of the melatonin metabolite, AFMK." J Pineal Res 45(4): 430-8. Manda, K., M. Ueno, et al. (2007). "Radiation-induced cognitive dysfunction and cerebellar oxidative stress in mice: protective effect of alpha-lipoic acid." Behav Brain Res 177(1): 7-14. Villasana, L., T. Pfankuch, et al. (2010). "Isoform-dependent effects of apoE on doublecortin- positive cells and microtubule-associated protein 2 immunoreactivity following (137)Cs irradiation." Radiat Environ Biophys 49(3): 421-6. Villasana, L., C. Poage, et al. (2008). "Passive avoidance learning and memory of 56Fe sham-irradiated and irradiated human apoE transgenic mice." Radiats Biol Radioecol 48(2): 167-70. Poulose, S. M., D. F. Bielinski, et al. (2011). "Exposure to 16O-particle radiation causes aging-like decrements in rats through increased oxidative stress, inflammation and loss of autophagy." Radiat Res 176(6): 761-9. Britten, R. A., L. K. Davis, et al. (2012). "Low (20 cGy) doses of 1 GeV/u (56)Fe-particle radiation lead to a persistent reduction in the spatial learning ability of rats." Radiat Res 177(2): 146-51. Whitley, R. J. (2002). "Herpes simplex virus infection." Semin Pediatr Infect Dis 13(1): 6-11.
Bibliography Type:	Description: (Last Updated: 11/12/2020)
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Abstracts for Journals and Proceedings	Sawtell NM, Williams MT, Wilson PF, Thompson RL. "Acute and long term outcomes of simulated deep space radiation exposure on latent viral CNS infection and CNS pathology." 41st International Herpes Virus Workshop, Madison, Wisconsin, July 23-27, 2016. 41st International Herpes Virus Workshop, Madison, Wisconsin, July 23-27, 2016. , Jul-2016
Abstracts for Journals and Proceedings	Doll J, Thompson RL, Williams MT, Sawtell NM. "Acute and long term outcomes of simulated deep space radiation exposure on latent viral CNS infection and CNS pathology." 37th Annual Graduate Student Research Forum, University of Cincinnati College of Medicine, Cincinnati, Ohio, November 21, 2016. 37th Annual Graduate Student Research Forum, University of Cincinnati College of Medicine, Cincinnati, Ohio, November 21, 2016. , Nov-2016